

from. These are very common in IEMs or In Ear Monitors which are used by musicians, singers and audio professionals to listen to a single channel of audio to base their work on the sound. For example, a singer in a band would use an IEM to listen to only the beats of the drummer so that he can base the singing on the beats. In this case, a stereo sound would ruin the experience. With the introduction of the stereo audio, a new dimension was added. Suddenly people discovered that not only could they get better audio quality, but also feel the direction in which the music was coming from. Stereo was largely popularised by The Beatles and the likes. Stereo sound or Stereophonic sound works by separating the sound into two separate channels, which are then directed to the right set of speakers. The recording of sound which is then mixed into stereo is particularly interesting. In traditional concert settings, the musicians are at different places and the audio is directed from the respective musician to the audience. But in the case of recording studios, the musicians are using microphones which are independent of where they are in the sound room. The direction of the music as perceived by the listener is actually the loudness of the sound. The closer the person is to a sound source, the louder the sound is. Hence to achieve stereo sound, one channel is louder in each earpiece, which gives an illusion of direction. In most of the songs by The Beatles, the instruments can be observed to be from the left channel while the voice is from the right channel. In case you remove one ear piece, you'd only hear the instruments or the voice prominently. But in most of the cases, you'll still hear the other channel very faintly. This is done to ensure that the music is loud enough when it reaches your ears.

While most ear pieces manufactured and sold today are stereo, there are still some mobile phone OEMs who bundle mono earphones to save costs. Some headphone manufacturers, especially for professional audio and gaming, bundle multiple drivers in the headphones to simulate multiple directions of sound. If you have headphones with 'surround' sound and multiple drivers, you'll be able to pin point the direction in which a player is sneaking up on you in games such as Counter Strike or even visualise the position of the musicians in an orchestra (if the audio is encoded that way).

Connectivity

One of the biggest classifications of headphones is in fact the medium of transmission of signals: wired or wireless.